

Features	Model	K-fold MAE	K-fold MSE	K-fold RMSE	K-fold R2	Y-test MAE	Y-test MSE	Y-test RMSE	Y-test R2
No AI, w/ WorkExp and JobSat	LinearRegression()	20568.543	780428037.998	27936.142	0.4673	17649.148	580915498.879	24102.189	0.5144
['WorkExp', 'JobSat', 'LearnCodeChoose', 'PurchaseInfluence', 'DatabaseChoice', 'PlatformChoice', 'WebframeChoice', 'DevEnvsChoice', 'Country', 'EdLevel', 'DevType', 'OrgSize', 'ICorPM', 'RemoteWork', 'Industry', 'LanguageChoice', 'lang_group_web_frontend', 'lang_group_data_science', 'lang_group_query_shell', 'lang_group_jvm', 'lang_group_dotnet']	DecisionTreeRegressor (max_depth=5, min_samples_leaf=5, random_state=42)	22095.447	923214331.061	30384.442	0.3648	19945.142	674469258.903	25970.546	0.4361
	DecisionTreeRegressor (max_depth=6, min_samples_leaf=10, random_state=42)	22159.146	915535823.658	30257.823	0.3824	20510.664	714025744.576	26721.260	0.4031
	Lasso(alpha=0.01)	21356.686	855166229.621	29243.225	0.4270	19322.806	663034235.707	25749.451	0.4457
	ElasticNet(alpha=0.01, l1_ratio=0.5)	20603.413	800044145.757	28285.052	0.4662	18449.144	624576077.467	24991.520	0.4779
	Ridge(alpha=1.0)	20466.111	772402416.450	27792.129	0.4741	17608.104	579920560.826	24081.540	0.5152
	HuberRegressor (epsilon=1.35, max_iter=1000)	20744.775	787328378.440	28059.372	0.4598	17807.791	595073762.434	24394.134	0.5025
	No AI, No WorkExp, no JobSat	LinearRegression()	22436.159	899651894.274	29994.198	0.3692	18053.130	586990313.702	24227.883
['LearnCodeChoose', 'PurchaseInfluence', 'DatabaseChoice', 'PlatformChoice', 'WebframeChoice', 'DevEnvsChoice', 'Country', 'EdLevel', 'DevType', 'OrgSize', 'ICorPM', 'RemoteWork', 'Industry', 'LanguageChoice', 'lang_group_web_frontend', 'lang_group_data_science', 'lang_group_query_shell', 'lang_group_jvm', 'lang_group_dotnet', 'lang_group_systems']	DecisionTreeRegressor (max_depth=5, min_samples_leaf=5, random_state=42)	24646.416	1103368043.582	33216.984	0.2256	22206.035	846077375.499	29087.409	0.2927
	DecisionTreeRegressor (max_depth=6, min_samples_leaf=10, random_state=42)	24430.588	1054146172.279	32467.617	0.2538	21385.931	769869672.405	27746.525	0.3564
	Lasso(alpha=0.01)	23167.379	975134003.080	31227.136	0.3260	20163.782	703686879.114	26527.097	0.4117
	ElasticNet(alpha=0.01, l1_ratio=0.5)	22287.670	901732010.835	30028.853	0.3738	19088.160	640708624.144	25312.223	0.4644
	Ridge(alpha=1.0)	22323.874	891221192.142	29853.328	0.3751	18032.938	585088912.539	24188.611	0.5109
	HuberRegressor (epsilon=1.35, max_iter=1000)	22677.946	915672306.498	30260.078	0.3555	18380.796	608179489.620	24661.295	0.4916
	With AI, WorkExp and JobSat	LinearRegression()	22779.342	1028937901.759	32077.062	0.3408	19630.759	952030080.651	30854.985
['WorkExp', 'JobSat', 'LearnCodeAI', 'LearnCodeChoose', 'PurchaseInfluence', 'DatabaseChoice', 'PlatformChoice', 'WebframeChoice', 'DevEnvsChoice', 'AIModelsChoice', 'AISelect', 'AIAgents', 'Country', 'EdLevel', 'DevType', 'OrgSize', 'ICorPM', 'RemoteWork']	DecisionTreeRegressor (max_depth=5, min_samples_leaf=5, random_state=42)	22348.738	939977932.832	30659.060	0.3577	20014.191	677772033.607	26034.055	0.4334
	DecisionTreeRegressor (max_depth=6, min_samples_leaf=10, random_state=42)	22405.664	929937911.922	30494.883	0.3744	20913.351	736869773.011	27145.345	0.3840
	Lasso(alpha=0.01)	21226.117	847924277.586	29119.139	0.4321	19354.662	662459422.547	25738.287	0.4462
	ElasticNet(alpha=0.01, l1_ratio=0.5)	20465.653	794187919.320	28181.340	0.4701	18560.616	623237857.099	24964.732	0.4790
	Ridge(alpha=1.0)	21344.379	831799941.813	28840.942	0.4368	18398.520	663605631.985	25760.544	0.4452

Features	Model	K-fold MAE	K-fold MSE	K-fold RMSE	K-fold R2	Y-test MAE	Y-test MSE	Y-test RMSE	Y-test R2		
'Industry', 'RemoteWork', 'LanguageChoice', 'AIThreat', 'AIComplex', 'AIAgentKnowledge', 'AIAgentOrchestration', 'AIAgentOrchestrationSecurity'	HuberRegressor (epsilon=1.35, max_iter=1000)	failed to converge									
	HuberRegressor (epsilon=1.35, max_iter= 5000)	22917.633	1011418344.535	31802.804	0.330	20085.386	1134258963.446	33678.761	0.052		
With AI, No WorkExp, No JobSat	LinearRegression()	24946.567	1235091085.093	35143.863	0.2344	20926.457	1095972523.768	33105.476	0.0838		
['LearnCodeAI', 'LearnCodeChoose', 'PurchaseInfluence', 'DatabaseChoice', 'PlatformChoice', 'WebframeChoice', 'DevEnvsChoice', 'AIModelsChoice', 'AISelect', 'AIAgents', 'Country', 'EdLevel', 'DevType', 'OrgSize', 'ICorPM', 'RemoteWork', 'Industry', 'LanguageChoice', 'AIThreat', 'AIComplex', 'AIAgentKnowledge']	DecisionTreeRegressor (max_depth=5, min_samples_leaf=5, random_state=42)	24798.750	1116959475.621	33420.944	0.2197	21713.381	811344696.178	28484.113	0.3217		
	DecisionTreeRegressor (max_depth=6, min_samples_leaf=10, random_state=42)	24452.378	1060200706.304	32560.723	0.2505	21279.302	755729531.884	27490.535	0.3682		
	Lasso(alpha=0.01)	23114.008	970306206.589	31149.738	0.3306	20147.473	704551284.495	26543.385	0.4110		
	ElasticNet(alpha=0.01, l1_ratio=0.5)	22217.539	895764289.664	29929.322	0.3803	19094.971	643069806.129	25358.821	0.4624		
	Ridge(alpha=1.0)	23349.006	966194780.505	31083.674	0.3382	19360.759	771285577.697	27772.029	0.3552		
	HuberRegressor (epsilon=1.35, max_iter=1000)	25209.744	1234665626.190	35137.809	0.2118	22007.281	1331290369.499	36486.852	-0.1130		

Models	Description	Best use case
LinearRegression()	It fits a linear model by minimizing residual sum of squares.	Strong linear relationship between features and target and features are not correlated
DecisionTreeRegressor(max_depth=5,min_samples_leaf=5,random_state=42)	A tree that recursively splits data to minimize variance in target values.	Non-linear relationships
Lasso(alpha=0.01)	Prevents over-fitting by automatically removing irrelevant features	Many features and not all are important
ElasticNet(alpha=0.01, l1_ratio=0.5)	mix of Ridge and Lasso. If l1_ratio is 1 it is like lasso, if 0, it acts more like Ridge	Usefull when features are many and correlated
Ridge(alpha=1.0)	Prevents over-fitting by penalizing large coefficients	data with correlated features
HuberRegressor(epsilon=1.35,max_iter=1000)	linear regressor that is less sensitive to outliers. sensitivity is set by epsilon	data with linear relationship but but with outliers
Findings:		
When All columns are not included, i.e. less features , Ridge has the highest R2 result in both Kfold and actual model.		
When All columns are included, ElasticNet (with l1_ratio=0.5) has the highest R2 score.		
Many of the columns probably correlate with each other		